

AI Driven Privacy Preserving Digital Ecosystem for Modernizing Sri Lanka's Paddy Supply Chain

R26-SE-007

Project Proposal Report

Chamudi K. S. I - IT22147950

B.Sc. (Hons) Degree in Information Technology Specialized in
Software Engineering

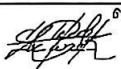
Department of Software Engineering

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Chamudi K. S. I.	IT22147950	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Dr. Mahima Weerasinghe	16/03/2026	

Name	Date	Signature
Mr. Eshan Weerasinghe	14/03/2026	

ABSTRACT

Sri Lanka's paddy supply chain remains largely dependent on informal communication channels, such as local collectors, phone calls, and messaging platforms, resulting in limited price transparency and inefficient market coordination. Farmers often lack access to reliable market information when deciding when and where to sell their harvest, while rice millers face difficulties in identifying trustworthy suppliers and estimating available supply. These limitations contribute to price uncertainty, unequal bargaining power, and reduced efficiency within the agricultural marketplace. Although several digital agriculture platforms have emerged globally, many existing solutions focus primarily on information dissemination and do not sufficiently address data privacy concerns or support intelligent price negotiation between stakeholders.

This research proposes an AI Powered Farmer Miller Marketplace as a key component of a broader AI driven privacy preserving digital ecosystem for modernizing Sri Lanka's paddy supply chain. The proposed system introduces an intelligent marketplace that connects farmers and rice millers while safeguarding sensitive user information and supporting fair price discovery. The marketplace integrates a Retrieval Augmented Generation (RAG) assistant to provide farmers with contextual market guidance using real agricultural data sources. A federated learning-based price prediction model is incorporated to estimate fair market prices without requiring centralized access to farmer data. Additionally, a secure matching mechanism prevents early exposure of farmer identities, reducing the risk of exploitation. The system also includes an AI-assisted negotiation module that helps both parties explore mutually beneficial price agreements.

The proposed platform aims to improve market transparency, empower farmers with informed decision-making capabilities, and facilitate more efficient supply matching for millers. By combining privacy-preserving machine learning with intelligent decision support, the research contributes toward building a trusted digital marketplace for Sri Lanka's agricultural sector.

Keywords: Paddy Supply Chain, Digital Agricultural Marketplace, Privacy Preserving Machine Learning, Federated Learning, AI Assisted Negotiation

ACKNOWLEDGEMENT

I would like to sincerely thank my supervisor, Dr. Mahima Weerasinghe, and my co-supervisor, Mr. Eishan Weerathna, for their exceptional guidance, encouragement, and unwavering support throughout the development of this research proposal. Their expertise in Artificial Intelligence and Software Engineering, coupled with their constructive feedback, has been invaluable in shaping the technical direction, clarity, and depth of this study.

I am also grateful to the Department of Software Engineering at the Sri Lanka Institute of Information Technology (SLIIT) for providing the academic resources, research facilities, and conducive environment that made this work possible.

Finally, I wish to acknowledge my colleagues, friends, and family for their constant encouragement, patience, and support, which have been a source of motivation and strength throughout this research journey. Their belief in my efforts has been instrumental in completing this proposal.

Table of Contents

DECLARATION	2
ABSTRACT	3
ACKNOWLEDGEMENT	4
LIST OF TABLES.....	7
LIST OF FIGURES	8
LIST OF ABBREVIATIONS.....	9
1.INTRODUCTION	10
1.1.Background and Context.....	10
1.2.Literature Review	12
Limitations of Existing Research and Systems - Summary	13
1.3.Research Gap	14
1.4.Research Problem	15
2.OBJECTIVE.....	16
2.1.Main Objective	16
2.2.Specific Objectives	16
3.METHODOLOGY	17
3.1.Research Approach	17
3.2.Technical Methods and Algorithms.....	18
RAG Based LLM Assistant for Market Advice.....	18
Federated Learning Simulation for Privacy-Preserving Price Prediction.....	18
Secure Matching Algorithm.....	19
Multi Agent Negotiation Mechanism.....	19
3.3Tools and Platforms with Justification.....	20
3.4Data Collection and Ethical Considerations	21
3.5Validation Strategy and Performance Metrics	21
Price Prediction Model Evaluation	21
Negotiation System Evaluation.....	21
Matching System Evaluation	22
Overall System Performance Evaluation	22
4.HIGH-LEVEL SYSTEM ARCHITECTURE	23
4.1.Overview of the Proposed Solution and Its Components.....	23
4.2.System Diagram of The Component.....	24
4.3.Individual Component: AI Powered Farmer Miller Marketplace	25
4.4.Component Interactions	27
4.5.Data Flow Explanation	27
4.6.Scalability and Security Considerations.....	28
5.USER REQUIREMENTS	29
5.1.Stakeholder Identification.....	29
Primary Stakeholders	29
Secondary Stakeholders	30
5.2.Functional Requirements	31
Paddy Listing and Marketplace.....	31
Secure Matching	31
AI Based Price Guidance	31

Multi Agent Negotiation	31
RAG Based LLM Advisory System.....	31
Transaction Management	31
5.3.Non-Functional Requirements	32
6.COMMERCIALIZATION PLAN.....	33
6.1.Target Market and Customer Persona.....	33
6.2.Value Proposition	33
6.3.Revenue Model	34
6.4.Cost Estimation.....	34
6.5.Pricing Strategy.....	35
6.6.Competitive Advantage	35
6.7.Intellectual Property Considerations	36
7.BUDGET AND JUSTIFICATION	36
7.1.Hardware Costs.....	36
7.2.Software Licenses	37
7.3.Cloud Services	37
7.4.Human Effort Cost.....	38
8.WORK BREAKDOWN STRUCTURE (WBS).....	39
9.GANTT CHART	40
10.REFERENCES LIST	41

LIST OF TABLES

Table 1 - List of Abbreviations	9
Table 2– Tools and platforms with justification	20
Table 3 - Haardware Cost	36
Table 4 - Software Cost	37
Table 5 - Cloud Service Cost	37
Table 6 - Human Effort Cost	38
Table 7 - Work Breakdown Structure	39

LIST OF FIGURES

Figure 1 - System Diagram	24
Figure 2 - System architecture Diagram of AI Powered Farmer Miller Marketplace	26
Figure 3 - Data Flow Diagram	28
Figure 4 - Gantt Chart	40

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
DFD	Data Flow Diagram
FAISS	Facebook AI Similarity Search
FL	Federated Learning
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing
PMB	Paddy Marketing Board
RAG	Retrieval Augmented Generation
REST	Representational State Transfer
UI	User Interface
UX	User Experience
DB	Database
SDG	Sustainable Development Goals
IoT	Internet of Things

Table 1 - List of Abbreviations

1. INTRODUCTION

1.1. Background and Context

Agriculture remains a fundamental pillar of Sri Lanka's economy, not only sustaining the nation's food security but also providing livelihoods for millions of rural households. Among agricultural crops, paddy cultivation occupies a central role, forming the backbone of rice production and serving as a primary source of income for smallholder farmers. Despite its importance, the journey of paddy from field to plate is riddled with inefficiencies. Farmers frequently navigate a fragmented marketplace, relying on informal networks, local middlemen, and phone calls to negotiate sales with rice millers. These traditional practices, while functional for decades, often leave farmers in vulnerable positions, exposed to opaque pricing and delayed market information.

The consequences of such a system are tangible. A farmer in Kurunegala or Anuradhapura may wake up to the uncertainty of whether the price offered by a miller is fair, or whether selling today versus waiting a week would yield higher income. Millers, on the other hand, struggle to predict supply and secure reliable paddy sources, leading to disruptions in production and unpredictable procurement costs. In both cases, decisions are made with incomplete information, leaving room for exploitation, financial loss, and systemic inefficiency. Globally, similar patterns are observed in agricultural markets where fragmented information flows hinder market efficiency, reduce farmer empowerment, and contribute to unstable commodity prices. In Sri Lanka, approximately 1.5 million farmers rely on paddy cultivation, yet studies indicate that nearly 60% of these farmers lack timely access to market information, highlighting a profound gap that directly affects income security and economic resilience.

Over the years, digital platforms and mobile applications have attempted to address these challenges by offering price listings, trade facilitation, and limited advisory services. While these solutions provide a basic layer of transparency, they fall short of meeting the complex needs of farmers and millers. Many platforms do not incorporate dynamic negotiation capabilities, leaving pricing rigid and unresponsive to market fluctuations. Others rely on central databases that store sensitive farmer data without adequate privacy protections, creating risks of misuse or exploitation. Moreover, these systems rarely contextualize recommendations based on seasonal trends, regional supply-demand variations, or government policy changes, making them less practical for farmers who must make timely decisions about selling their harvest.

Recognizing these gaps, this research proposes an AI-Powered Farmer–Miller Marketplace with Privacy & Negotiation, a digital ecosystem designed to transform paddy trading in Sri Lanka. The system integrates advanced AI technologies with privacy-preserving mechanisms to empower farmers, optimize miller procurement, and ensure equitable trade. At its core, the marketplace features a Retrieval-Augmented Generation (RAG) assistant, which provides personalized, actionable guidance to farmers. By analyzing historical price trends, government policies, and local supply-demand dynamics, the RAG assistant helps farmers make informed decisions about when and where to sell their paddy, reducing uncertainty and maximizing potential income.

To protect farmer data, the system leverages federated learning, a machine learning approach that allows predictive models to be trained across distributed devices without centralizing sensitive information. This ensures that farmers' personal and yield data remain confidential

while still enabling accurate forecasting of fair market prices. In addition, a secure matching mechanism anonymizes farmer and miller identities during initial interactions, safeguarding against exploitation and creating a more balanced negotiation environment. Finally, a multi-agent AI negotiation module supports both parties in exploring optimal pricing strategies. By simulating potential negotiation outcomes and providing algorithmically guided suggestions, the module facilitates fairer transactions that reflect real-world market conditions, benefiting both farmers and millers.

The significance of this research extends beyond operational efficiency. By fostering transparent, data-driven interactions and equitable market access, the proposed system directly contributes to the United Nations Sustainable Development Goals (SDGs), particularly SDG 2 (Zero Hunger) and SDG 8 (Decent Work and Economic Growth). Farmers gain improved income security and confidence in market participation, while millers benefit from predictable supply and reduced procurement risk. These outcomes collectively strengthen rural economies, support sustainable agriculture, and promote food security at a national level.

From a research perspective, this project aligns with the AI and Software Engineering research cluster, combining elements of machine learning, distributed computing, human-computer interaction, and privacy-preserving technologies. By integrating AI-driven decision support with secure, interactive negotiation tools, the system represents a practical application of intelligent software design in a real-world agricultural context. It also exemplifies how digital innovation can address socio-economic challenges, translating technical research into tangible benefits for communities.

In essence, the AI-Powered Farmer Miller Marketplace seeks to bridge the gap between tradition and technology, providing smallholder farmers with the tools to participate confidently in modern markets, while ensuring that millers can procure paddy efficiently and fairly. By embedding intelligence, security, and user-centric design into the trading process, the marketplace offers a holistic, implementable, and socially impactful solution to longstanding inefficiencies in Sri Lanka's paddy supply chain. Beyond Sri Lanka, the framework and methodologies developed through this research have the potential to inspire scalable AI-driven solutions in other agricultural sectors and emerging economies facing similar market fragmentation and information asymmetry.

1.2.Literature Review

Recent advancements in artificial intelligence (AI), machine learning (ML), and data-driven technologies have significantly influenced research in modern agricultural systems. Researchers increasingly explore the potential of intelligent technologies to address challenges such as agricultural decision support, market inefficiencies, data privacy, and digital trade facilitation. Several studies investigate technological approaches relevant to this research, including retrieval augmented generation (RAG), federated learning, AI-based advisory systems, and digital agricultural marketplaces. This section reviews key scholarly contributions in these areas, evaluates existing systems and platforms, and highlights limitations that motivate the development of the proposed AI Powered Farmer Market Marketplace with Privacy and Negotiation.

One growing research direction involves the development of AI driven advisory systems that assist farmers in making informed agricultural decisions. Recent progress in large language models and retrieval techniques has enabled systems capable of generating context aware recommendations based on domain specific knowledge sources.

Sawant proposed a retrieval augmented generation (RAG) framework that integrates semantic search with large language models to deliver contextual agricultural guidance to farmers, enabling them to access expert knowledge related to crop cultivation and management practices [1].

Similarly, **AgroLLM** demonstrates how domain specific knowledge bases combined with RAG architectures can provide accurate and contextually relevant responses to agricultural queries [2]. These systems show that RAG-based approaches can significantly enhance knowledge accessibility for farmers. However, most existing advisory systems primarily focus on agronomic decision making, such as crop management and disease control, rather than market related decision support including pricing strategies, buyer selection, and trading recommendations.

Another critical research area concerns privacy preserving machine learning in agricultural data ecosystems. As digital agriculture platforms collect increasing amounts of farm level data, concerns regarding data privacy and ownership have become increasingly important. Federated learning (FL) has emerged as a promising approach for distributed machine learning, enabling collaborative model training without requiring raw data to be centralized. **Kairouz et al.** present a comprehensive overview of federated learning architectures and highlight its effectiveness in protecting sensitive data across distributed participants [3].

Similarly, **Zalik and Zalik** review the applications of federated learning in agriculture, including crop prediction and smart farming analytics, demonstrating its ability to preserve farmer data confidentiality while enabling collective model improvements [4].

Additional studies emphasize the advantages of federated learning in distributed environments such as smart agriculture, where farm data originates from geographically dispersed locations [5]. Despite its benefits, federated learning also introduces challenges such as communication overhead, heterogeneous datasets, and limited computing resources in rural environments. These issues highlight the need for carefully designed architectures when implementing privacy preserving machine learning in agricultural systems.

Beyond advisory systems and privacy preserving analytics, a significant body of literature focuses on the development of digital technologies for smart agriculture and agricultural supply chains. **Research on smart farming ecosystems** demonstrates how technologies such as the Internet of Things (IoT), data analytics, and machine learning can improve crop productivity and resource efficiency [6].

Wolfert et al. discuss how data driven agriculture platforms can enhance decision making by integrating multiple stakeholders across agricultural supply chains [7].

Furthermore, **studies on AI enabled agricultural supply chain management** explore the use of intelligent algorithms to improve coordination between producers, distributors, and buyers, ultimately increasing supply chain efficiency and reducing transaction costs [8]. While these systems demonstrate the transformative potential of digital technologies in agriculture, most research primarily focuses on production optimization, crop monitoring, or logistics management rather than direct marketplace interactions between farmers and buyers.

In addition to academic research, several **digital marketplace platforms and agricultural trading applications** have been introduced to connect farmers with buyers. These systems provide services such as price listings, market information dashboards, and basic trading interfaces. Research examining digital agricultural marketplaces indicates that such platforms can improve transparency and reduce information asymmetry between producers and buyers [9]. However, many of these platforms still rely on centralized data architectures and static price listings. They rarely provide intelligent decision support mechanisms that assist farmers in determining optimal selling times, negotiating prices, or identifying suitable buyers based on dynamic market conditions.

Another emerging research area involves the use of **AI driven negotiation and multi agent systems** in digital marketplaces. Multi agent negotiation frameworks allow autonomous agents to represent different stakeholders and simulate bargaining strategies in electronic markets. Studies on automated negotiation systems demonstrate that AI agents can assist participants in reaching mutually beneficial agreements by analyzing preferences, constraints, and historical transaction data [10]. Although such systems have been explored in domains such as e-commerce and supply chain management, their application within agricultural marketplaces remains limited.

Limitations of Existing Research and Systems - Summary

Overall, the literature demonstrates significant progress in the development of AI technologies for agriculture, particularly in advisory systems, privacy preserving machine learning, and digital supply chain platforms. Nevertheless, several limitations remain.

Existing advisory systems mainly address crop production guidance rather than market decision support. Privacy preserving techniques such as federated learning have not been widely applied in agricultural marketplaces. Furthermore, digital agricultural platforms typically lack intelligent negotiation mechanisms that support fair price discovery between farmers and buyers. These limitations highlight the need for an integrated digital marketplace that combines AI based market guidance, privacy preserving analytics, and negotiation support.

The proposed **AI Powered Farmer Miller Marketplace with Privacy and Negotiation** seeks to address these limitations by integrating retrieval augmented advisory capabilities, federated learning-based price prediction, secure matching mechanisms, and AI assisted negotiation support into a unified marketplace platform tailored for the Sri Lankan paddy supply chain.

1.3. Research Gap

Although recent research has made significant progress in applying artificial intelligence and digital technologies to agriculture, several important gaps remain in the development of intelligent, privacy preserving agricultural marketplaces. The literature review highlights advancements in AI driven advisory systems, federated learning for distributed machine learning, and digital agricultural supply chain platforms. However, these research efforts typically address these technologies independently rather than integrating them into a unified ecosystem that supports market driven decision making for farmers.

First, many **AI-based agricultural advisory systems** focus primarily on production-oriented recommendations, such as crop selection, disease detection, irrigation management, and yield prediction. While these systems provide valuable support for farm management, they rarely address market related decision making, which is a critical factor influencing farmer income. Farmers often lack access to intelligent tools that help them determine optimal selling times, estimate fair market prices, or identify the most suitable buyers. As a result, farmers particularly smallholder farmers remain vulnerable to information asymmetry and market exploitation by intermediaries.

Second, existing digital agricultural marketplaces generally function as simple trading platforms that display product listings and buyer offers. Although such systems improve connectivity between farmers and buyers, they typically lack advanced AI driven decision support capabilities. Most platforms do not provide predictive pricing models, automated negotiation support, or intelligent matching between farmers and millers based on quality, location, and demand conditions. Consequently, these systems fail to fully leverage the potential of artificial intelligence to create fair and efficient agricultural markets.

Another major limitation concerns **data privacy and data ownership**. Many digital agriculture platforms rely on centralized data collection architectures where farmer data is stored and processed by a central authority. This approach raises concerns regarding data privacy, misuse of sensitive agricultural information, and unequal control over data resources. While federated learning has been proposed as a promising privacy preserving solution for distributed machine learning, its application within agricultural marketplaces particularly for price prediction and market analytics remains largely unexplored in current research.

Furthermore, **automated negotiation mechanisms** have been widely studied in electronic commerce and multi agent systems, where AI agents assist participants in reaching optimal agreements. However, the application of AI-based negotiation techniques within agricultural trading platforms is still limited. Most existing agricultural marketplaces rely on static pricing or manual negotiation processes, which often disadvantage small scale farmers who lack market knowledge or bargaining power.

These limitations indicate the absence of an integrated digital solution that simultaneously addresses **market intelligence, privacy protection, and negotiation support** within agricultural trading systems. In particular, the Sri Lankan paddy supply chain faces challenges related to price transparency, intermediary dominance, and limited digital infrastructure for direct farmer buyer interaction. Current technological solutions do not adequately support farmers in making informed market decisions while maintaining control over their sensitive agricultural data.

Therefore, there is a clear need for an intelligent digital marketplace that combines **AI-driven market guidance, privacy preserving machine learning, and automated negotiation capabilities** within a unified platform. The proposed AI Powered Farmer Miller Marketplace with Privacy & Negotiation aims to fill this research gap by integrating retrieval augmented advisory systems, federated learning-based price prediction, secure buyer seller matching, and AI assisted negotiation mechanisms. By addressing these gaps, the system seeks to empower farmers with improved market access, enhanced price transparency, and stronger bargaining capabilities while ensuring the privacy and ownership of their agricultural data.

1.4. Research Problem

Despite the increasing adoption of digital technologies in agriculture, many farmers, especially smallholder farmers continue to face significant challenges in accessing fair and transparent markets for selling their produce. In the Sri Lankan paddy supply chain, farmers often rely on intermediaries or local traders to sell their harvest, which frequently leads to price manipulation, limited bargaining power, and lack of transparency in the trading process. As a result, farmers may receive lower profits compared to the actual market value of their produce.

One major issue contributing to this problem is the lack of intelligent digital platforms that support direct and efficient interactions between farmers and millers. Existing digital agricultural platforms primarily provide basic market information or listing services, but they do not offer advanced decision support tools that help farmers determine optimal selling strategies or negotiate better prices. Without such support, farmers remain disadvantaged when engaging in trade negotiations with more experienced buyers.

Another critical challenge involves data privacy and trust in digital agricultural systems. Many modern digital platforms rely on centralized data storage where sensitive farmer information such as production volumes, pricing data, and transaction records is stored in centralized databases. This raises concerns about data ownership, privacy, and potential misuse of agricultural data. Farmers may therefore be reluctant to share information that could otherwise improve market predictions and decision-making systems.

Additionally, most agricultural trading systems lack automated mechanisms for fair price negotiation and intelligent matching between buyers and sellers. Manual negotiation processes can be inefficient and biased, particularly when farmers have limited access to market intelligence or historical price trends. Without intelligent support systems, farmers struggle to make informed decisions about when to sell their products and how to negotiate effectively.

Therefore, the key research problem addressed in this study is the **lack of an intelligent, privacy preserving digital marketplace that supports direct interaction, market intelligence, and fair negotiation between farmers and millers in the paddy supply chain**. Addressing this problem requires the development of an AI driven system capable of providing

market insights, protecting sensitive data, and facilitating efficient trading interactions. The proposed AI Powered Farmer Miller Marketplace with Privacy and Negotiation aims to address these challenges by integrating advanced AI techniques with secure digital marketplace technologies to create a more transparent, fair, and efficient trading environment for agricultural stakeholders.

2. OBJECTIVE

2.1.Main Objective

The main objective of this research is to design and develop an AI-powered digital marketplace that enables privacy-preserving, intelligent, and transparent interactions between farmers and millers in the Sri Lankan paddy supply chain.

2.2.Specific Objectives

1. Develop a Digital Marketplace Platform

To develop a digital marketplace platform that facilitates direct trading between paddy farmers and rice millers. The platform will allow farmers to list details of their harvest, including quantity, location, and quality attributes, enabling millers to discover available paddy supplies efficiently. By providing a centralized digital interface, the system aims to reduce reliance on intermediaries and improve market accessibility for farmers. This functionality will enhance transparency and enable more efficient interactions between buyers and sellers.

2. Implement an AI-Based Market Intelligence Module

To implement an AI-based market intelligence module that analyzes historical price data, seasonal trends, and regional demand patterns. This module will provide farmers with data driven recommendations regarding optimal selling prices and potential trading opportunities. By transforming complex market data into actionable insights, the system will support informed decision-making for farmers. Ultimately, this component aims to reduce price uncertainty and improve farmers' ability to maximize their profits.

3. Design a Privacy Preserving Mechanism

To design and implement a privacy preserving price prediction mechanism using federated learning techniques. This approach allows predictive models to be trained collaboratively using distributed farmer data while ensuring that sensitive information remains locally stored and protected. By avoiding centralized data collection, the system enhances user trust and safeguards confidential agricultural data. This mechanism will enable accurate price forecasting without compromising farmer privacy.

4. Integrate an AI Assisted Negotiation Support System

To integrate an AI assisted negotiation support system that helps farmers and millers engage in fair and informed price discussions. The system will analyze historical transaction data and market trends to recommend appropriate price ranges and negotiation strategies. Through algorithmically guided insights, both parties can better understand reasonable price expectations before finalizing transactions. This component aims to promote balanced bargaining power and fair price discovery in the marketplace.

5. Implement Intelligent Buyer Seller Matching Algorithms

To implement intelligent matching algorithms that recommend suitable trading partners based on factors such as geographic proximity, paddy quantity, quality specifications, and buyer demand. These algorithms will improve the efficiency of the marketplace by connecting farmers with millers who best match their trading requirements. By automating the partner discovery process, the platform can reduce search time and improve transaction success rates. This functionality contributes to a more streamlined and responsive digital trading environment.

6. Evaluate the Effectiveness of the Proposed Marketplace

To evaluate the effectiveness of the proposed AI-powered marketplace in improving price transparency, trading efficiency, and decision-making support for farmers. The evaluation will involve analyzing system performance, user interactions, and transaction outcomes after implementation. Metrics such as price fairness, trading speed, and user satisfaction will be considered during assessment. The results will help determine the practical impact and scalability of the proposed system within the paddy trading ecosystem.

3. METHODOLOGY

3.1. Research Approach

This research follows a Design Science Research (DSR) approach to design, develop, and evaluate an intelligent digital marketplace that addresses inefficiencies and privacy concerns in the Sri Lankan paddy supply chain. Design Science is appropriate for this study because the primary goal is to create and evaluate an innovative technological artifact that solves a real-world problem.

The proposed artifact is an AI-powered marketplace platform that connects farmers and millers while ensuring fair pricing, privacy protection, and intelligent decision support. The system integrates multiple AI technologies such as Retrieval Augmented Generation (RAG) based Large Language Models, Federated Learning, Secure Matching algorithms, and Multi-Agent Negotiation mechanisms.

The research process follows the typical Design Science phases:

1. **Problem Identification**
Analyze issues in the paddy supply chain such as price exploitation, lack of transparency, and data privacy concerns.
2. **Objective Definition**
Define system goals including fair price discovery, privacy preserving data usage, and intelligent advisory services.
3. **Artifact Design and Development**
Design and implement the AI-powered marketplace system architecture and its components.

4. **Demonstration**
Simulate the system using prototype applications for farmers and millers.
5. **Evaluation**
Assess system performance using predefined metrics such as negotiation success rate, prediction accuracy, and system efficiency.

This approach ensures that the proposed system is both technically innovative and practically applicable to real agricultural contexts.

3.2. Technical Methods and Algorithms

The proposed marketplace integrates several advanced AI and privacy preserving techniques to enable intelligent and secure market interactions.

RAG Based LLM Assistant for Market Advice

The marketplace will include a Retrieval-Augmented Generation (RAG) based Large Language Model (LLM) assistant to provide contextual market advice to farmers.

This module combines a language model with an external knowledge base containing agricultural reports, historical market prices, and paddy trading information. When a farmer asks a question, the system retrieves relevant documents using **vector similarity search** and supplies them as contextual information to the language model.

The model then generates a response that provides recommendations such as estimated price ranges, selling suggestions, and negotiation guidance.

The RAG pipeline consists of the following steps:

1. User query submission from the farmer application.
2. Conversion of the query into embeddings.
3. Retrieval of relevant market information from a vector database.
4. Generation of a contextual response using the LLM.

This approach enables the system to deliver accurate, context-aware agricultural market guidance.

Federated Learning Simulation for Privacy-Preserving Price Prediction

To protect sensitive agricultural data, the system adopts a federated learning-based price prediction model. In this approach, machine learning models are trained across distributed data sources without sharing raw data with a central server.

Since implementing a large-scale federated network is complex, the research will implement a **federated learning simulation** where multiple local datasets represent different market participants such as farmers, regional markets, and millers.

Each local dataset trains a model independently, and the model parameters are then aggregated to create a global price prediction model.

This approach ensures that sensitive information remains private while enabling collaborative learning for accurate market predictions.

Secure Matching Algorithm

The system includes a secure matching mechanism to connect farmers with potential millers while protecting participant identities during the early stages of negotiation.

The matching algorithm evaluates several attributes including:

- Paddy type
- Quantity available
- Expected price range
- Geographic location

A weighted scoring function is used to identify the most suitable matches between buyers and sellers. To maintain privacy, the system assigns anonymous identifiers to participants so that identities are revealed only after a negotiation agreement is reached.

This mechanism prevents early exposure of farmer identities and reduces the risk of unfair market influence.

Multi Agent Negotiation Mechanism

To support fair price discovery, the platform incorporates a multi agent negotiation system where software agents represent both farmers and millers during the bargaining process.

The negotiation model follows a rule-based bargaining strategy, where each agent gradually adjusts its price offer based on predefined negotiation parameters. For example, the farmer agent may reduce the initial price by a small percentage in each negotiation round, while the miller agent gradually increases its offer.

The negotiation process continues until the price difference between the agents falls within an acceptable threshold, at which point an agreement is reached.

This automated negotiation mechanism helps simulate fair bargaining conditions and reduces power imbalances between trading participants.

3.3 Tools and Platforms with Justification

The prototype system will be implemented using modern software development tools and AI frameworks that support rapid development and experimentation.

Tool / Platform	Purpose	Justification
Python	AI and machine learning development	Python provides extensive libraries for implementing AI models, federated learning simulations, and data processing tasks.
Node.js	Backend API and marketplace services	Node.js enables scalable server-side development and efficient handling of real-time marketplace interactions between farmers and millers.
LangChain	RAG pipeline implementation	Simplifies integration between LLMs and external knowledge sources for generating contextual market advice.
OpenAI API / LLaMA Models	Language model implementation	Enables intelligent conversational assistance for farmers through natural language understanding and generation.
FAISS / ChromaDB	Vector database for semantic search	Efficient similarity search for RAG retrieval
Scikit-learn	Machine learning models	Useful for price prediction and data analysis experiments
Flower Framework	Federated learning simulation	Enables experimentation with distributed model training
MongoDB	Data storage	Provides flexible NoSQL storage for marketplace transactions, negotiation data, and user profiles.
React Native	Prototype user interface	Enables development of cross-platform mobile applications for farmers and millers to interact with the marketplace system.

Table 2– Tools and platforms with justification

These technologies provide a scalable and flexible development environment suitable for implementing the proposed research prototype.

3.4 Data Collection and Ethical Considerations

The research requires market related data such as historical paddy prices, trading patterns, and agricultural market reports. Data will be obtained from publicly available sources including agricultural research publications, government datasets, and collaboration with the regional PMB office.

In cases where real world data is limited, synthetic datasets and simulated marketplace transactions will be generated to support experimentation and model evaluation.

Ethical considerations are an important part of this research. The following measures will be implemented:

- Protecting farmer data through federated learning techniques
- Avoiding centralized storage of sensitive agricultural information
- Using anonymized or simulated data where necessary
- Ensuring transparency in AI generated market recommendations
- Designing negotiation algorithms that promote fairness and avoid bias

These practices ensure responsible use of data and protect stakeholder privacy.

3.5 Validation Strategy and Performance Metrics

The effectiveness and reliability of the proposed AI-Powered Farmer Miller Marketplace will be evaluated through systematic experimental testing of each major system component. The validation strategy focuses on quantitative and qualitative performance metrics to assess accuracy, efficiency, and user satisfaction.

Price Prediction Model Evaluation

The price prediction module, developed using federated learning, will be evaluated for its predictive accuracy. The following metrics will be used:

- Mean Absolute Error (MAE): Measures the average absolute difference between predicted and actual paddy market prices. A lower MAE indicates higher prediction accuracy.
- Root Mean Squared Error (RMSE): Evaluates the square root of the average squared differences between predicted and actual prices, penalizing larger errors more heavily than MAE.

These metrics will provide a clear assessment of the model's ability to forecast fair market prices and support informed decision making for farmers.

Negotiation System Evaluation

The AI assisted negotiation engine will be assessed using the following metrics:

- **Negotiation Success Rate:** Percentage of negotiation sessions that successfully reach an agreement between farmers and millers.
- **Average Number of Negotiation Rounds:** Measures the efficiency of the negotiation process by recording the number of iterations required to reach a mutually acceptable price.
- **Fairness of Final Price:** Compares the agreed transaction price with predicted market prices to ensure equitable outcomes for both parties.

These metrics will quantify the negotiation engine's effectiveness in facilitating fair and efficient price discovery.

Matching System Evaluation

The intelligent buyer seller matching component will be evaluated using:

- **Matching Accuracy:** Proportion of recommended trading partners that result in successful transactions.
- **Relevance of Recommendations:** Degree to which suggested buyers or sellers meet the farmer's or miller's requirements (e.g., location, quantity, and quality).
- **Average Time to Identify Matches:** Measures the efficiency of the system in suggesting suitable trading partners.

This evaluation will demonstrate the algorithm's ability to connect stakeholders effectively, reducing search time and improving trading efficiency.

Overall System Performance Evaluation

In addition to component level validation, the overall system will be assessed under realistic operational scenarios to evaluate:

- **Response Time of AI Services:** Time taken by the RAG assistant, price prediction model, and negotiation engine to process requests and provide outputs.
- **Scalability:** System performance when handling multiple simultaneous users and concurrent transactions.
- **User Interaction Efficiency:** Measures the ease and speed with which users (farmers and millers) can perform key tasks within the marketplace, based on simulated trading scenarios.

The results from these evaluations will collectively demonstrate the feasibility, reliability, and practical effectiveness of the proposed AI-powered marketplace. They will provide evidence of the system's ability to enhance **price transparency, trading efficiency, and decision-making support** for farmers and millers alike.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1. Overview of the Proposed Solution and Its Components

The proposed research introduces an **AI driven, privacy preserving digital ecosystem designed to modernize Sri Lanka's paddy supply chain** by integrating intelligent analytics, secure data management, and automated decision support technologies. The ecosystem aims to address several long-standing challenges in the agricultural sector, including limited access to reliable market intelligence, inefficient coordination among stakeholders, lack of predictive insights for farmers, and concerns related to transparency and data privacy.

To overcome these limitations, the proposed architecture combines artificial intelligence models, distributed learning mechanisms, secure digital infrastructure, and intelligent marketplace services into a unified platform. The architecture follows a layered design, where data collection, data processing, AI intelligence, and user interaction operate as separate but interconnected system layers. This modular approach allows the system to efficiently process large volumes of agricultural and market data while ensuring scalability and security.

The overall ecosystem consists of four major AI-driven components developed as part of the collaborative research project:

1. Digital Twin for Paddy Farming

This component creates a continuously updated virtual representation of paddy fields by integrating multimodal agricultural data such as climate conditions, soil characteristics, and farm metadata. Advanced computer vision models and graph neural networks are used to analyze crop health and environmental interactions.

2. AI Driven Paddy Price Forecasting

This component predicts short term and long-term paddy price trends using transformer based time series models combined with causal machine learning methods. The system analyzes historical prices, climate conditions, production patterns, and economic indicators to generate explainable market predictions.

3. Blockchain-Based Warehouse Coordination

This module focuses on improving warehouse management and stock transparency through blockchain technology and privacy-preserving cryptographic mechanisms. It enables reliable tracking of warehouse inventory while supporting disaster-aware logistics coordination.

4. AI Powered Farmer Miller Marketplace

This component enables secure digital trading between farmers and rice millers. It integrates privacy preserving federated learning, intelligent buyer seller matching, automated negotiation agents, and AI based decision support to ensure fair price discovery and efficient marketplace transactions.

Together, these components form an integrated ecosystem that combines AI intelligence, secure data sharing, and intelligent decision support mechanisms to improve efficiency, transparency, and sustainability within Sri Lanka's paddy supply chain.

4.2. System Diagram of The Component

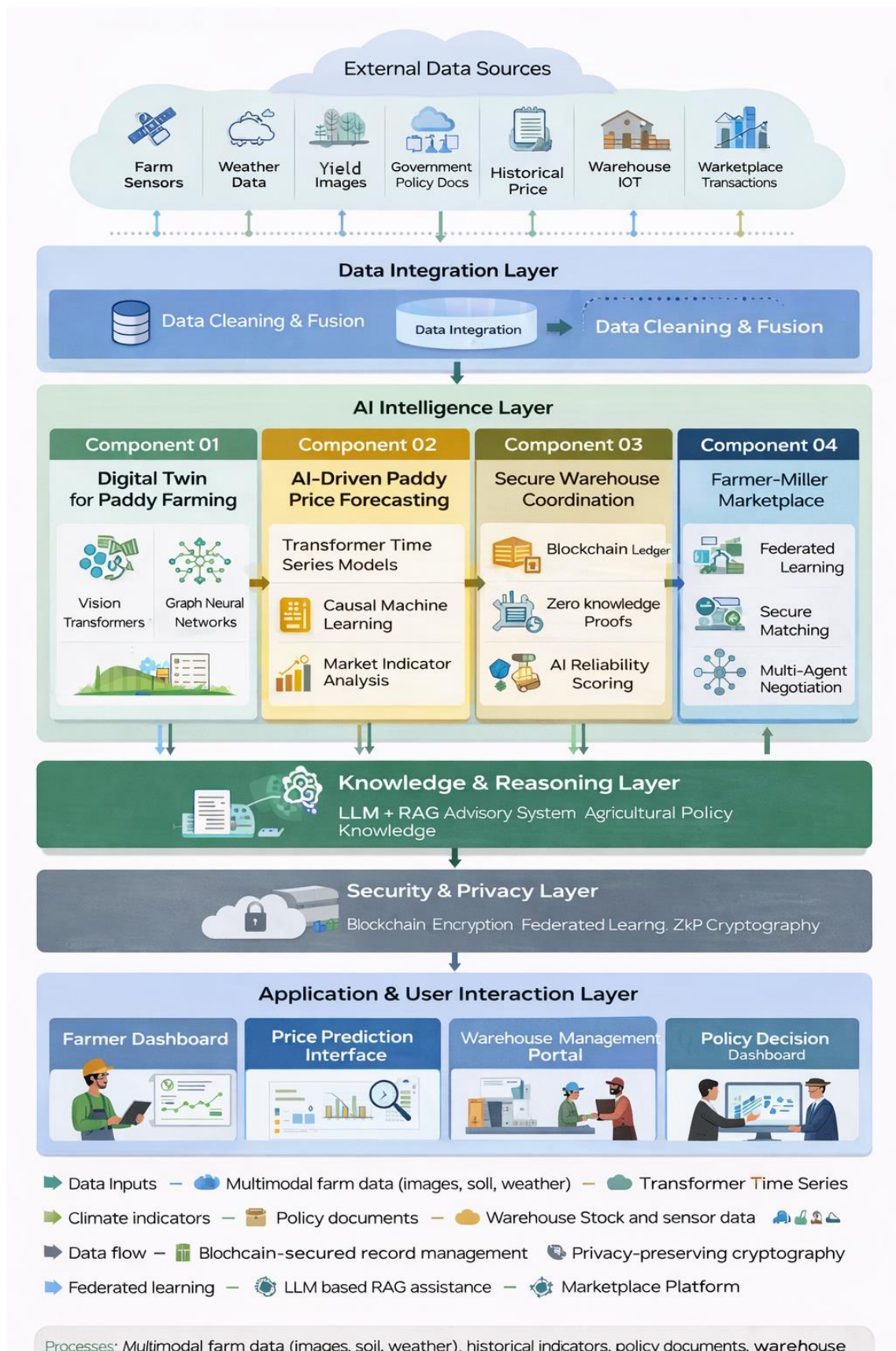


Figure 1 - System Diagram

The **overall system architecture diagram** illustrates how the four research components interact within the proposed digital ecosystem. The diagram presents multiple architectural layers including external agricultural data sources, data integration processes, AI intelligence modules, security mechanisms, and user interaction interfaces.

At the top of the architecture are **external data sources**, which include farm sensors, weather data, historical market prices, government policy documents, warehouse data, and marketplace transaction records. These heterogeneous data sources provide the raw information required for agricultural monitoring, price prediction, and supply chain management.

The collected data flows into the **data integration layer**, where data cleaning, transformation, and fusion processes combine information from multiple sources into structured datasets. These datasets are then processed by the **AI intelligence layer**, which contains the four major system components developed within the research project.

Below the intelligence layer is the **knowledge and reasoning layer**, where a Retrieval Augmented Generation (RAG) based large language model analyzes policy documents, agricultural reports, and market knowledge to generate contextual advisory insights for system users.

The architecture also includes a **security and privacy layer** that ensures data protection through mechanisms such as blockchain encryption, federated learning, and privacy preserving cryptographic protocols.

Finally, the **application and user interaction layer** provides dashboards and mobile interfaces through which farmers, warehouse managers, policymakers, and market participants interact with the system.

4.3.Individual Component: AI Powered Farmer Miller Marketplace

While the overall ecosystem integrates multiple components, this research specifically focuses on the AI Powered Farmer Miller Marketplace, which enables intelligent and secure trading interactions between farmers and rice millers.

The marketplace component is designed to address several challenges in traditional agricultural markets, including information asymmetry, unfair price negotiation, and lack of transparency in buyer seller interactions. The system introduces an AI assisted digital marketplace where farmers can list paddy harvests, discover potential buyers, and negotiate prices using intelligent support tools.

The architecture of this component consists of several core modules:

RAG Based LLM Market Advisory System

Provides contextual market advice to farmers by retrieving information from agricultural knowledge sources and generating responses using a large language model.

Federated Learning Simulation Module

Enables privacy preserving machine learning by training price prediction models on distributed datasets without sharing raw user data.

Secure Matching Engine

Matches farmers and millers based on attributes such as paddy type, quantity, location, and expected price while protecting participant identities during early interactions.

Multi Agent Negotiation System

Simulates bargaining between buyers and sellers using AI agents that dynamically adjust price offers during negotiation rounds.

These modules collectively support intelligent trading interactions, fair price discovery, and privacy protection within the digital marketplace.



Figure 2 - System architecture Diagram of AI Powered Farmer Miller Marketplace

4.4.Component Interactions

The different components within the ecosystem interact through a coordinated data exchange process that supports agricultural analytics, supply chain coordination, and marketplace transactions.

Agricultural data collected through the **Digital Twin component** contributes insights about crop production levels and farm conditions. These insights help estimate supply availability within the market.

At the same time, the **AI Driven Price Forecasting component** analyzes historical market prices, climate conditions, and economic indicators to generate predicted price ranges for paddy. These predicted prices are shared with the marketplace component and serve as reference indicators for fair price negotiation.

The **Secure Warehouse Coordination component** maintains real-time information about warehouse capacity and stock availability. This information supports logistical planning and ensures that traded paddy can be stored and transported efficiently.

The Farmer Miller Marketplace component integrates information from these components to support intelligent trading decisions. Farmers receive market insights and predicted price ranges, while millers are matched with suitable sellers based on supply availability and market demand.

4.5.Data Flow Explanation

The data flow within the proposed system begins with the collection of agricultural, market, and environmental data from multiple external sources. These data sources include farm sensors, weather reports, historical market prices, and warehouse monitoring systems. The collected data is transmitted to the data integration layer, where it is cleaned, transformed, and combined into structured datasets.

The integrated datasets are then processed by the AI intelligence layer, where different components perform specialized analysis tasks. The Digital Twin component analyzes agricultural conditions to estimate production levels and crop health, while the AI-Driven Price Forecasting component processes historical market data and economic indicators to predict future paddy price trends.

The predicted price ranges generated by the price forecasting system are transmitted to the AI Powered Farmer Miller Marketplace, where they are used as reference values for negotiation and decision support. When a farmer lists paddy for sale through the marketplace interface, the listing data is stored in the marketplace database and processed by the secure matching engine, which identifies suitable millers based on trading requirements.

Once a match is identified, the multi agent negotiation system initiates a bargaining process between the farmer and miller agents. During this process, farmers can consult the RAG based LLM advisory system to obtain recommendations based on agricultural policies, market reports, and historical trading patterns.

Transaction data generated from marketplace interactions is continuously fed back into the system, contributing to improved price prediction models and market analysis. This cyclical data flow ensures that the ecosystem continuously learns from market activity and adapts to changing supply chain conditions.

Figure 3 - Data Flow Diagram

The proposed architecture is designed to support both scalability and strong security guarantees. The modular architecture allows each component to operate as an independent service, enabling the system to scale horizontally as the number of users and data sources increases. AI models, marketplace services, and data processing modules can be deployed as separate microservices, allowing computational workloads to be distributed across cloud infrastructure. Security and privacy are key design priorities within the ecosystem. Sensitive agricultural and transaction data are protected using federated learning mechanisms, which prevent the sharing of raw user data with centralized servers. Blockchain-based encryption and privacy-preserving cryptographic protocols ensure transparency and integrity within warehouse and supply chain operations.

5. USER REQUIREMENTS

5.1.Stakeholder Identification

The proposed AI powered farmer miller marketplace operates within the broader paddy supply chain ecosystem in Sri Lanka and involves several stakeholders who interact with the system in different capacities. Identifying these stakeholders is important to ensure that the platform effectively supports the needs of all users involved in paddy production, trading, and market analysis. The stakeholders can be categorized into primary stakeholders, who directly interact with the system, and secondary stakeholders, who benefit indirectly from the insights and data generated by the platform.

Primary Stakeholders

Primary stakeholders are the users who directly engage with the marketplace system to perform transactions, manage data, or maintain the platform.

Paddy Farmers

Paddy farmers represent the main suppliers in the marketplace and play a central role in the system. Farmers use the platform to create listings for their harvested paddy, specifying details such as quantity, location, and expected price. Through the AI powered negotiation system, farmers can interact with potential buyers and receive price recommendations based on predicted market trends. The integration of the RAG based advisory system further supports farmers by providing guidance on pricing strategies, market conditions, and agricultural policies. By using this platform, farmers can reduce dependency on intermediaries and gain access to a more transparent and competitive marketplace.

Rice Millers

Rice millers act as the primary buyers in the marketplace. They use the platform to browse available paddy listings, evaluate potential suppliers, and negotiate purchasing prices with farmers. The system helps millers efficiently identify suitable suppliers based on factors such as paddy quantity, geographic location, and predicted market prices. The automated negotiation mechanism enables millers to reach mutually beneficial agreements while minimizing time spent in manual bargaining processes. As a result, the platform improves procurement efficiency and enhances market transparency for millers.

Paddy Marketing Board (PMB)

The Paddy Marketing Board (PMB) is a government organization responsible for stabilizing paddy prices and supporting farmers within the national agricultural sector. Within the proposed system, PMB may act as a monitoring and regulatory stakeholder. The organization can use aggregated marketplace data to observe market trends, price fluctuations, and regional supply patterns. Such insights can assist the PMB in making informed decisions related to price control policies, government purchasing programs, and national food security planning. The presence of PMB as a stakeholder also contributes to improving trust and transparency within the digital marketplace.

System Administrators

System administrators are responsible for managing and maintaining the technical infrastructure of the platform. Their responsibilities include managing user accounts, ensuring system security, monitoring platform performance, and maintaining the reliability of backend services. Administrators also oversee data management processes and ensure that the platform operates smoothly without interruptions. By maintaining the integrity and availability of the system, administrators play a critical role in ensuring that all marketplace operations function effectively.

Secondary Stakeholders

Secondary stakeholders do not directly participate in marketplace transactions but benefit from the insights, analytics, and data generated by the system.

Agricultural Analysts and Researchers

Agricultural analysts and researchers can use aggregated data generated by the marketplace to study patterns in paddy production, price fluctuations, and market behavior. Such data can contribute to academic research, agricultural planning, and the development of improved farming strategies. Insights derived from marketplace transactions may also support the design of better agricultural policies and data-driven decision-making.

Government Policy Makers

Government policy makers may benefit from the system by analyzing nationwide trends related to paddy supply, demand, and pricing. The data generated through the marketplace can provide valuable insights for developing policies related to agricultural subsidies, food security, and rural economic development.

5.2.Functional Requirements

Functional requirements describe the core features and operations that the system must provide to support the marketplace and AI based decision support.

Paddy Listing and Marketplace

FR3: The system shall allow farmers to create paddy listings including quantity, location, and expected price.

FR4: The system shall store and manage all marketplace listings in the marketplace database.

FR5: The system shall allow millers to browse and search available paddy listings.

Secure Matching

FR6: The system shall match farmers with potential millers based on location, quantity requirements, and trading preferences.

FR7: The system shall ensure that user identities remain hidden during the initial matching stage to protect privacy.

AI Based Price Guidance

FR8: The system shall integrate predicted paddy price ranges obtained from the AI driven price forecasting component.

FR9: The system shall display price prediction insights to assist farmers and millers during negotiations.

Multi Agent Negotiation

FR10: The system shall support automated negotiation between farmer agents and miller agents to determine fair transaction prices.

FR11: The system shall allow both parties to accept, reject, or counter proposed offers during negotiation.

RAG Based LLM Advisory System

FR12: The system shall provide an AI advisory assistant that answers farmer questions.

Transaction Management

FR14: The system shall allow farmers and millers to confirm final trade agreements.

FR15: The system shall store completed transactions in a transaction database for record keeping and analysis.

5.3.Non-Functional Requirements

Non-functional requirements define the quality attributes of the system, including security, performance, and usability.

Security

The system must ensure strong data protection and privacy for all users. Sensitive user data should be protected using secure authentication mechanisms and encrypted communication protocols. The platform must also support privacy-preserving techniques such as secure matching and federated learning to prevent unauthorized access to sensitive trading information.

Performance

The system should provide efficient responses to user requests, including marketplace searches and negotiation interactions. AI models used for price prediction and advisory services should deliver results within acceptable response times to ensure smooth user experience.

Scalability

The architecture should be designed to support an increasing number of farmers and millers without performance degradation. The system should allow easy scaling of backend services and databases as marketplace usage grows.

Usability

The platform should provide a simple and intuitive interface for farmers and millers. Since some users may have limited technical experience, the system must prioritize ease of navigation, clear instructions, and user-friendly interactions.

Reliability

The system should maintain stable operation and ensure that marketplace data, negotiation processes, and transaction records remain consistent and available even during high usage periods.

6. COMMERCIALIZATION PLAN

6.1. Target Market and Customer Persona

The proposed AI powered farmer miller marketplace primarily targets participants within the Sri Lankan paddy supply chain, particularly paddy farmers and rice millers who are directly involved in the buying and selling of paddy harvests. The platform is designed to improve market transparency, reduce dependency on intermediaries, and support fair price discovery through intelligent digital technologies.

The primary customer persona is **small and medium scale paddy farmers**, who often face challenges in identifying reliable buyers and negotiating fair market prices. These farmers typically rely on traditional trading methods or local intermediaries, which can limit their bargaining power and access to accurate market information. By using the proposed system through a mobile application, farmers can list their harvest, view predicted price ranges, and negotiate directly with millers.

The second important customer persona is **rice millers**, who require consistent access to paddy supply for rice production. Millers can use the platform to identify available paddy listings, compare supplier options, and negotiate prices efficiently through the automated negotiation system.

In addition to these direct users, government organizations such as the **Paddy Marketing Board (PMB)** represent institutional stakeholders who may utilize aggregated system insights to monitor market trends, regional supply patterns, and price fluctuations.

The initial target market for system deployment would be selected paddy producing regions in Sri Lanka.

6.2. Value Proposition

The proposed system offers several key benefits that address existing inefficiencies within the paddy supply chain.

First, the platform enables direct digital connectivity between farmers and millers, reducing reliance on intermediaries and enabling more transparent trading interactions. This can improve price fairness and increase farmer profitability.

Second, the integration of AI driven price prediction models provides users with insights into expected market price ranges. This information supports better decision-making during negotiation and helps reduce uncertainty in paddy trading.

Third, the secure matching and privacy preserving mechanisms ensure that user identities and sensitive trading information are protected during early negotiation stages. This improves trust among participants and encourages wider adoption of the platform.

Finally, the inclusion of a RAG based advisory assistant enables farmers to obtain instant guidance related to market conditions, agricultural policies, and pricing strategies, helping them make informed selling decisions.

Together, these capabilities create a digital ecosystem that improves efficiency, transparency, and fairness within the paddy trading process.

6.3.Revenue Model

Since the proposed AI Powered Farmer Miller Marketplace is primarily developed as a research oriented system aimed at improving agricultural market transparency, the platform is not designed to generate direct commercial revenue during the research phase. Instead, the focus is on creating a sustainable and accessible digital infrastructure that benefits farmers and rice millers.

During the initial stages, the system is expected to operate as a free digital service for farmers, ensuring that smallholder farmers can access market information, trading opportunities, and AI-assisted decision support without financial barriers. This approach is particularly important in rural agricultural communities where technology adoption may be limited by affordability.

In the long term, the sustainability of the platform may rely on institutional or government support, particularly from agricultural authorities such as the Paddy Marketing Board or the Department of Agriculture, which may utilize the platform to improve market transparency and monitor paddy supply trends. These institutions could support the platform through funding, partnerships, or integration with existing agricultural information systems.

6.4.Cost Estimation

The primary costs associated with implementing the system include software development, infrastructure, and operational expenses.

Development costs involve building the mobile application interface, backend APIs, AI model integration, and database systems. These components can be developed using widely available open-source frameworks such as Node.js, Python, and React Native, which helps minimize development expenses.

Infrastructure costs include cloud hosting services required to run the backend services, databases, and AI models. These costs may involve cloud platforms such as AWS, Google Cloud, or similar infrastructure providers.

Additional operational costs include system maintenance, security monitoring, data storage, and periodic model updates to maintain prediction accuracy.

For the initial research prototype, infrastructure costs can remain relatively low by deploying the system using lightweight cloud services and open source tools.

6.5.Pricing Strategy

The pricing strategy should prioritize affordability and adoption, particularly for small scale farmers who represent the majority of users in the paddy market.

The marketplace platform itself can remain free for farmers, encouraging widespread adoption and participation in the digital trading ecosystem.

Revenue generation can instead focus on transaction-based fees charged to millers, who generally have greater purchasing capacity. These fees can remain small so that they do not discourage platform usage.

Institutional users such as the Paddy Marketing Board or agricultural research organizations may access premium analytical dashboards or market intelligence reports through subscription-based pricing models.

This approach ensures that the system remains inclusive while still maintaining financial sustainability.

6.6.Competitive Advantage

The proposed system provides several advantages compared to traditional agricultural marketplaces and existing digital trading platforms.

First, the system incorporates multi-agent negotiation mechanisms, allowing automated negotiation between farmers and millers to discover mutually beneficial prices more efficiently.

Second, the use of privacy preserving techniques such as federated learning and secure matching protects user data and improves trust among marketplace participants.

Finally, the integration of a RAG based LLM advisory assistant provides farmers with contextual guidance based on agricultural reports, policies, and historical market data. This feature transforms the platform from a simple trading system into an intelligent decision-support ecosystem.

These technological innovations collectively create a platform that is more intelligent, transparent, and user-focused than traditional paddy trading mechanisms.

6.7. Intellectual Property Considerations

The proposed system may involve several intellectual property considerations related to the software architecture, AI models, and marketplace algorithms.

The research outputs, including the AI models, negotiation mechanisms, and system architecture, may be documented and published as part of academic research contributions. These contributions can demonstrate the novelty of integrating AI-based price forecasting, federated learning, and negotiation agents within a digital agricultural marketplace.

If the system is further developed into a commercial product, certain components such as the negotiation algorithms, AI advisory mechanisms, and system integration architecture could potentially be protected through software copyrights or intellectual property registration.

In addition, partnerships with organizations such as the Paddy Marketing Board may require agreements regarding data usage, system ownership, and deployment rights, ensuring that sensitive agricultural data is handled responsibly.

7. BUDGET AND JUSTIFICATION

The development and evaluation of the proposed AI powered farmer miller marketplace require several resources including computing hardware, software tools, cloud infrastructure, and human development effort. Since the project is primarily software based, most costs are associated with cloud computing resources and system development activities. The estimated budget focuses on resources required to implement a research prototype and conduct system evaluation.

7.1. Hardware Costs

The project does not require specialized hardware equipment because the development and testing can be conducted using existing personal computers and mobile devices owned by the research team members. These devices are sufficient for software development, AI model training, and system testing.

However, minimal hardware resources may be required for system demonstrations and testing, including mobile devices for validating the marketplace application.

Item	Estimated Cost (LKR)
Developer Laptop (existing resource)	0
Android Mobile Device for Testing	0
Internet Connectivity (research period)	20,000
Total Hardware Cost	20,000

Table 3 - Hardware Cost

Justification

Existing computing devices are sufficient for development and testing activities. A mobile device is required to test the mobile interface of the marketplace system and ensure usability for farmers. Internet connectivity costs are included to support continuous development, cloud deployment, and testing activities during the research period.

7.2. Software Licenses

Most of the technologies used in the project are open source, which significantly reduces software licensing costs. The system can be developed using open-source programming frameworks and development tools.

Software Tool	Estimated Cost (LKR)
Programming frameworks (Python, Node.js, React)	0
Database systems (PostgreSQL / MongoDB)	0
Development tools (VS Code / GitHub)	0
AI frameworks (TensorFlow / PyTorch)	0
Total Software License Cost	0

Table 4 - Software Cost

Justification

Open-source development tools and frameworks provide sufficient capabilities for implementing AI models, backend services, and marketplace interfaces without requiring additional licensing fees. This approach significantly reduces project expenses while maintaining flexibility for development and experimentation.

7.3. Cloud Services

Cloud infrastructure is required to host backend services, databases, and AI models used by the system. Cloud platforms provide scalable computing resources necessary for running the AI models and supporting user interactions within the marketplace.

Cloud Service	Estimated Cost (LKR)
Database hosting (MongoDB Free Tier)	0
AI model API Access	9,000
Total Cloud Service Cost	9,000

Table 5 - Cloud Service Cost

Justification

Cloud infrastructure allows the system to operate as a platform accessible by farmers and millers from different locations. It also enables scalable deployment of AI models and ensures reliable data storage for marketplace transactions and prediction datasets.

7.4. Human Effort Cost

Although the research project is conducted by university students, it requires significant time and effort for system design, software development, AI model implementation, testing, and documentation. Estimating human effort provides an understanding of the true development cost if the system were implemented commercially.

Role	Estimated Cost (LKR)
AI Model Development	0
Backend Development	0
Frontend Development	0
System Testing and Integration	0
Total Human Effort Cost	0

Table 6 - Human Effort Cost

Overall Justification

The estimated budget reflects the resources required to design, develop, and evaluate the proposed AI powered farmer miller marketplace as a research prototype. The project leverages open-source technologies and existing hardware resources to minimize costs while still enabling the development of advanced AI driven functionalities. Cloud infrastructure provides scalable deployment and supports system accessibility for potential users. The inclusion of human effort cost highlights the development effort required to implement the system in a real-world environment.

8. WORK BREAKDOWN STRUCTURE (WBS)

Task / Component	Description	Timeline
Requirement Analysis	Gather stakeholder requirements, identify functional and non-functional needs, and finalize scope.	Weeks 1–3
System Architecture Design	Design overall system architecture, AI modules, data flow, and component interactions.	Weeks 2–4
RAG-Based LLM Advisory Module	Implement retrieval-augmented generation model for market guidance; connect knowledge base.	Weeks 3–7
Federated Learning Price Prediction	Develop privacy-preserving price prediction models using distributed data; implement simulations.	Weeks 4–8
Secure Matching Engine	Design algorithms for matching farmers and millers based on quantity, location, and price.	Weeks 5–8
Multi-Agent Negotiation System	Implement AI agents to simulate negotiations; optimize fair price discovery mechanisms.	Weeks 6–10
Marketplace Frontend Development	Develop mobile app interface for farmers and millers using React Native.	Weeks 4–12
Backend & API Development	Build server-side APIs, authentication, and integration with AI modules.	Weeks 4–12
System Integration	Integrate frontend, backend, AI modules, and secure data flows into a cohesive system.	Weeks 10–12
Testing & Validation	Conduct component-level and end-to-end testing; evaluate metrics for prediction, negotiation, and matching.	Weeks 12–13
Documentation & Report Writing	Prepare technical report, user manuals, and academic documentation.	Weeks 11–12

Table 7 - Work Breakdown Structure

9. GANTT CHART

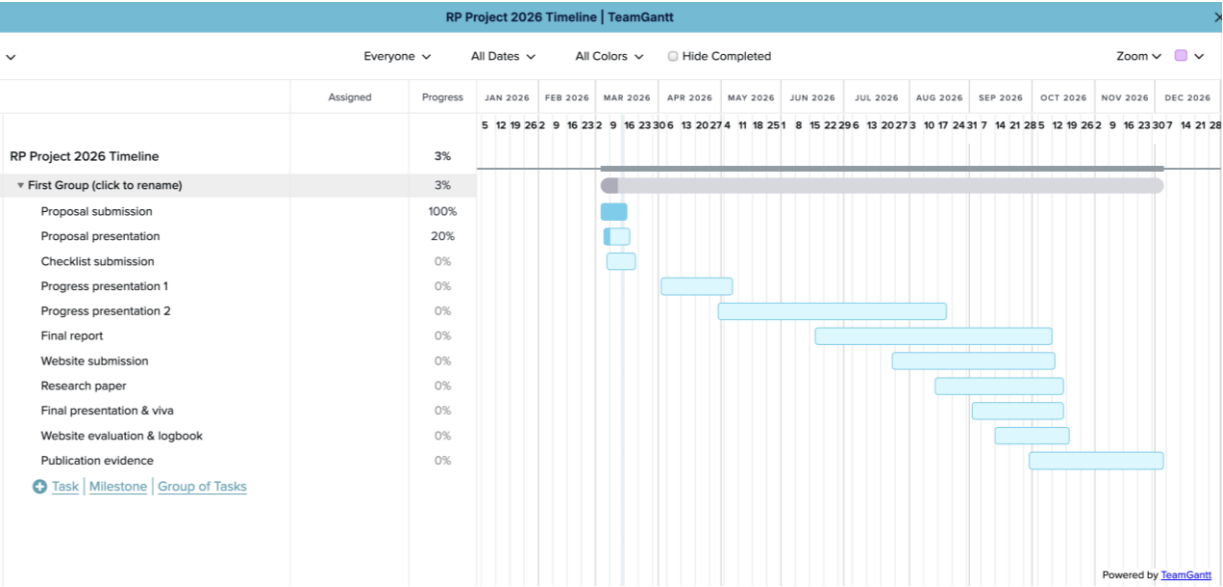


Figure 4 - Gantt Chart

10. REFERENCES LIST

- [1] S. Sawant, “Retrieval-Augmented Generation for Agricultural Advisory Systems,” 2023. [Reference 1](#)
- [2] J. Li et al., “AgroLLM: Large Language Models for Agricultural Knowledge Assistance,” 2023. [Reference 2](#)
- [3] P. Kairouz et al., “Advances and Open Problems in Federated Learning,” *Foundations and Trends in Machine Learning*, 2021. [Reference 3](#)
- [4] B. Žalik and M. Žalik, “Federated Learning Applications in Smart Agriculture,” 2022. [Reference 4](#)
- [5] H. Yang et al., “Federated Machine Learning: Concept and Applications,” *ACM Transactions*, 2019. [Reference 5](#)
- [6] J. Wolfert et al., “Big Data in Smart Farming: A Review,” *Agricultural Systems*, 2017. [Reference 6](#)
- [7] M. Kamilaris and F. X. Prenafeta-Boldú, “Deep Learning in Agriculture,” *Computers and Electronics in Agriculture*, 2018. [Reference 7](#)
- [8] S. Ivanov et al., “AI in Supply Chain Management,” *International Journal of Production Research*, 2019.
- [9] T. Reardon et al., “Digital Technologies in Agricultural Value Chains,” *World Development*, 2021.
- [10] J. Jennings et al., “Automated Negotiation in Multi-Agent Systems,” *Artificial Intelligence*, 2001.